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The Editor-in-Chief
Universal Access in the Information Society, Springer Nature

Dear Editor,

We submit the research article *"Detecting Text-Reading Hallucinations in Vision-Language Descriptions of Cyrillic Signage, Packaging and Documents for Blind Users"* by Rostyslav Zatserkovnyi, Roksoliana Zatserkovna and Petro Kutsyk for consideration in *Universal Access in the Information Society*.

Context. Reading printed text remains among the most common reasons blind people take photographs, as the VizWiz-era studies of their visual questions showed (Bigham et al., UIST 2010; Brady et al., CHI 2013). General-purpose vision-language models now do much of this reading in scene-description apps. When such a model cannot read a word, it tends to produce a plausible one instead: a familiar brand in place of an unusual one, a complete street name where the sign is cut off, or a date that is one digit off. Blind users have few ways to notice this and tend to trust automatic descriptions (MacLeod et al., CHI 2017), so a wrong expiry date, dosage or amount due is an accessibility failure. We study Ukrainian (Cyrillic) text, which is under-represented in accessibility research.

What we did. We measure how often four models from a single vendor (Anthropic's Claude family) assert text that is not printed, and whether this can be flagged at run time. A controlled study of 567 synthetic images varies image quality, framing and word type. A second study of 176 real images (Open Food Facts packaging and official Ukrainian documents) uses ground truth independent of the models. Every quoted word is scored deterministically, and two automated raters from different vendors audit the labeller.

Findings. Results are given as ranges across the four models. Only 2.8–5.6% of quoted words are hallucinated, yet 35–73% of descriptions contain at least one such word, mostly without a hedge. The main mechanism is normalising unusual printed words to familiar ones: when one letter on real document pages was edited to create a near-word, the models reported the real word it resembles on 44–56% of pages. Errors specific to the Ukrainian letters і, ї, є, ґ do occur, but on real text they are only 2–3% of hallucinated words, so we present Ukrainian as the setting rather than as the main source of risk. Combining a second reading by the model, an independent OCR pass and agreement between models separates hallucinated from correct words with an area under the ROC curve (AUROC) of 0.86–0.97, where 1 is perfect and 0.5 is chance. With the cheapest model, a description plus a second reading costs about US$0.0024 per real image, with median latencies of 8–10 s per call.

Fit with the journal. The work concerns equal access to printed information for blind and visually impaired people through technology already in everyday use. It gives developers of assistive reading apps an inexpensive safeguard: telling users which words to doubt. The study covers one model family and includes no user study. Whether such flags help blind users is the next question, and we release the benchmarks, ground truth and code to support it.

The manuscript is original, has not been published previously and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests. The use of generative AI is disclosed in the manuscript's declarations.

Sincerely,

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